Assignment 9:-

1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE odate = '1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.183 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
d2_96712_Devanand> ALTER TABLE Orders
    -> ADD CONSTRAINT unique_onum UNIQUE (onum);
Query OK, 0 rows affected (0.707 sec)
Records: 0  Duplicates: 0  Warnings: 0

d2_96712_Devanand> CREATE UNIQUE INDEX idx_onum
    -> ON Orders(onum);
Query OK, 0 rows affected, 1 warning (0.356 sec)
Records: 0  Duplicates: 0  Warnings: 1

d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE odate = '1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.014 sec)
```

3. Create an index that would permit salesperson to retrieve his orders.

```
d2_96712_Devanand> CREATE INDEX idx_salesman
    -> ON Orders(snum);
Query OK, 0 rows affected (0.532 sec)
Records: 0  Duplicates: 0  Warnings: 0

d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE snum = 1001;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.010 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
d2_96712_Devanand> INSERT INTO Customers (cname, snum, rating) VALUES ('A', 1001, 5);
Query OK, 1 row affected (0.069 sec)

d2_96712_Devanand> INSERT INTO Customers (cname, snum, rating) VALUES ('B', 1002, 5);
Query OK, 1 row affected (0.026 sec)

d2_96712_Devanand>
d2_96712_Devanand> -- Not Allowed (same snum + rating)
Query OK, 0 rows affected (0.002 sec)

d2_96712_Devanand> INSERT INTO Customers (cname, snum, rating) VALUES ('C', 1001, 5);
Query OK, 1 row affected (0.140 sec)

d2_96712_Devanand>
d2_96712_Devanand> SELECT * FROM CUSTOMERS;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2001 | Hoffman  | London   |    100 | 1001 |
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2006 | Clemens  | London   |    100 | 1001 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
| 2007 | Pereira  | Rome     |    100 | 1004 |
| NULL | A        | NULL     |      5 | 1001 |
| NULL | B        | NULL     |      5 | 1002 |
| NULL | C        | NULL     |      5 | 1001 |
+------+----------+----------+--------+------+
10 rows in set (0.009 sec)
```

5. Create an index to speed up searching order on a given date by given customer.

```
d2_96712_Devanand> CREATE INDEX idx_orderdate_cust
    -> ON Orders(Odate, cnum);
Query OK, 0 rows affected (0.705 sec)
Records: 0  Duplicates: 0  Warnings: 0

d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE odate = '1990-10-03'
    -> AND cnum = 2001;
+------+--------+------------+------+------+
| onum | amt    | odate      | cnum | snum |
+------+--------+------------+------+------+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
+------+--------+------------+------+------+
1 row in set (0.130 sec)
```